3.1.2 The digestive system provides an interface with the environment. Digestion involves enzymic hydrolysis producing smaller molecules that can be absorbed and assimilated.

The digestive system

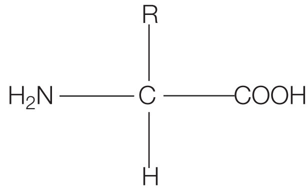
The gross structure of the human digestive system limited to oesophagus, stomach, small and large intestines and rectum. The glands associated with this system limited to the salivary glands and the pancreas.

Digestion is the process in which large molecules are hydrolysed by enzymes to produce smaller molecules that can be absorbed and assimilated.

Proteins

Proteins have a variety of functions within all living organisms.

The general structure of an amino acid as



Condensation and the formation of peptide bonds linking together amino acids to form polypeptides. The relationship between primary, secondary, tertiary and quaternary structure, and protein function.

The biuret test for proteins.

Enzyme action

Enzymes as catalysts lowering activation energy through the formation of enzyme-substrate complexes.

The lock and key and induced fit models of enzyme action.

Enzyme properties

The properties of enzymes relating to their tertiary structure. Description and explanation of the effects of temperature, competitive and non-competitive inhibitors, pH and substrate concentration.

Candidates should be able to use the lock and key model to explain the properties of enzymes. They should also recognise its limitations and be able to explain why the induced fit model provides a better explanation of specific enzyme properties.

Carbohydrate digestion

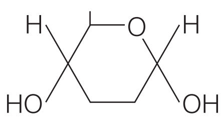
Within this unit, carbohydrates should be studied in the context of the following

- starch, the role of salivary and pancreatic amylases and of maltase located in the intestinal epithelium
- disaccharides, sucrase and lactase.

Biological molecules such as carbohydrates and proteins are often polymers and are based on a small number of chemical elements.

Monosaccharides are the basic molecular units (monomers) of which carbohydrates are composed.

The structure of α-glucose as



α-glucose

and the linking of α-glucose by glycosidic bonds formed by condensation to form maltose and starch.

Sucrose is a disaccharide formed by condensation of glucose and fructose.

Lactose is a disaccharide formed by condensation of glucose and galactose.

Lactose intolerance.

Biochemical tests using Benedict's reagent for reducing sugars and non-reducing sugars. Iodine/potassium iodide solution for starch.